YAZAN ELTOBGY

Automation Engineer | Robotics Engineer | AI Engineer

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Automation and Robotics Engineer combining industrial automation (PLC, PROFINET, MES, OPC-UA, IIoT) with autonomous robotics (ROS/ROS2, navigation, SLAM, sensor fusion). Leads an Industry 4.0 smart factory lab and has built award-winning autonomous robots. Published researcher in IT/OT integration and energy optimization.

EXPERIENCE

RA Automation Engineer — Research & Development
Feb 2026 – Present

German Jordanian University (GJU)

- Leading the Festo CP Factory Industry 4.0 lab as sole technical lead; commissioned the ASRS32-P station (4-axis PPPR, Siemens ET200SP, PROFINET, RFID) and operating the RASS robot station (Mitsubishi RV-4FL with 3-gripper auto-change)
- Configured MES4 and integrated it with CIROS for end-to-end production orders; built a Plant Simulation model validated within 6% of measured throughput, identifying the robot assembly station as the bottleneck
- Co-authored two papers: Applied Sciences (IT/OT integration, MES-based control) and an accepted ICFT 2026 paper (energy optimization, 47% lower energy per batch from one to three carriers)

Robotics & AI Engineer (Part-Time)
Feb 2026 – Sep 2026

KOUN Technologies, Vienna

- Developed WALLIE warehouse droid: ROS2/Gazebo simulation and URDF with Fusion 360 mesh
- Implemented PPRPP kinematics and camera-based gripper calibration
- Implemented YOLO-based package detection and state machine (LOCKED/FREE/LOST/DISABLED) for autonomous multi-SKU handling

Supervisor Engineer — Al Fahd Quadruped Robot
2025 – 2026

Al Hussein Technical University (HTU)

- Ported full ROS2 stack (URDF, launch files, configs) and integrated ros2_control for real-time joint control
- Achieved stable locomotion on real hardware — resolved all real-robot implementation gaps
- 1st Place — Best Graduation Project at HTU

Autonomous Service Robot Research
2024 – June 2025

Graduation Project — The Hashemite University

- Developed hybrid navigation system combining reinforcement learning with traditional algorithms
- Implemented full sensor fusion (camera, LiDAR, encoders) with SLAM mapping on Raspberry Pi
- National Winner — Best Graduation Project in Robotics and AI (2025)

Robotics Research Intern
Jul 2024 – Oct 2024

Karunya Institute of Technology and Sciences, India

- Published: 'Autonomous Mobile Robot Navigation using RL with Nav2'
- Optimized ROS2 navigation stack reducing path planning computation time by 25%
- Developed reinforcement learning algorithms for dynamic environment navigation

Industrial IoT & Automation Intern
Aug 2023 – Jan 2024

INJO4.0, Jordan

- Implemented predictive maintenance systems reducing equipment downtime by 40%
- Automated data collection workflows processing 5000+ daily sensor readings
- Developed real-time monitoring dashboards using PLCs and Norvi IIoT devices

EDUCATION

Bachelor of Science in Mechatronics Engineering
The Hashemite University, Zarqa, Jordan | September 2025

SKILLS

Automation:	PLC Programming, TIA Portal, PROFINET, OPC-UA, Modbus, MQTT, SCADA, MES, IoT/IIoT, Industry 4.0
Robotics:	ROS/ROS2, Gazebo, Isaac Sim, SLAM, Nav2, ros2_control, CARLA
Programming:	C++, Python, CUDA, MATLAB
AI / ML:	Machine Learning, Deep Learning, Reinforcement Learning, OpenCV, TensorFlow, PyTorch, YOLO (v8)
Hardware:	Arduino, Raspberry Pi, Jetson Nano, ESP32, STM32
Design / Tools:	Tecnomatix Plant Simulation, SolidWorks, AutoCAD, Fusion 360, Blender, GitHub, Linux, Docker, n8n

PROJECTS

CP Factory Industry 4.0 Research
Tecnomatix Plant Simulation, TIA Portal, PROFINET
Festo CP Factory digital twin; energy & predictive-maintenance research. Published in Applied Sciences; energy paper accepted at ICFT 2026.

7-DOF Industrial Arm Robot
ROS2, MoveIt2, YOLO
Simulation-to-hardware 7-DOF robotic arm controlled with MoveIt2 and ROS2, with YOLO-based perception for object detection and task execution.

Autonomous Vehicle — CARLA
CARLA, ROS2, Computer Vision
Urban autonomous driving: traffic recognition & obstacle avoidance in simulation.

Mobile Robot RL Navigation
RL, TurtleBot3, LiDAR
Published research on RL-based autonomous navigation with Nav2.

AI Smart Radar System
YOLOv8, DeepSORT, OCR
Dual-camera speed violation detection, vehicle tracking & plate recognition.

Cancer Detection — Deep Learning
TensorFlow, PyTorch
High-accuracy cancer detection from medical images with inference API.